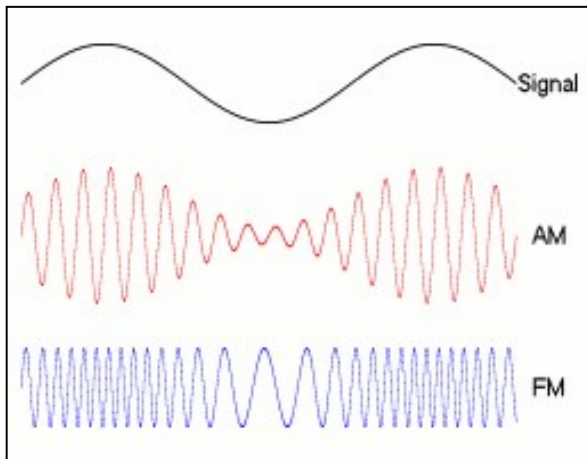


ECE 3317

Dr. Jiefu Chen

Notes 9

Transmission Lines (Frequency Domain)

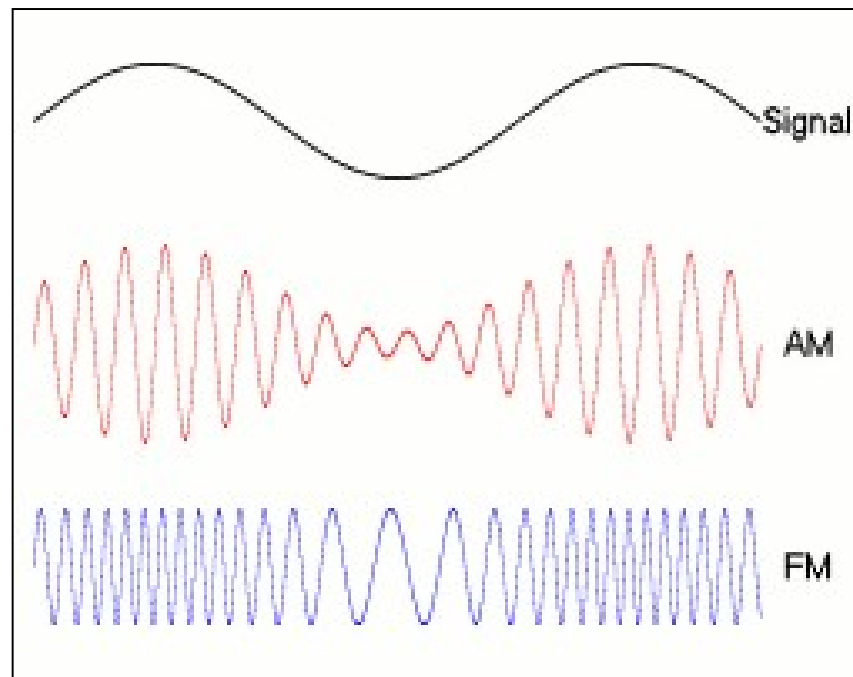


Adapted from notes by Prof. Stuart A. Long

Frequency Domain

Why is the frequency domain important?

- Most communication systems use a sinusoidal signal (which may be modulated).



Frequency Domain (cont.)

Why is the frequency domain important?

- A solution in the frequency-domain allows to solve for an arbitrary time-varying signal on a lossy line (by using the Fourier transform method).

Fourier-transform pair

$$\begin{aligned}\tilde{V}(\omega) &= \int_{-\infty}^{\infty} V(t) e^{-j\omega t} dt \\ V(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \tilde{V}(\omega) e^{j\omega t} d\omega\end{aligned}$$

For a physically-realizable (real-valued) signal, we can write

$$V(t) = \frac{1}{\pi} \int_0^{\infty} \text{Re}\left\{ \tilde{V}(\omega) e^{j\omega t} \right\} d\omega$$

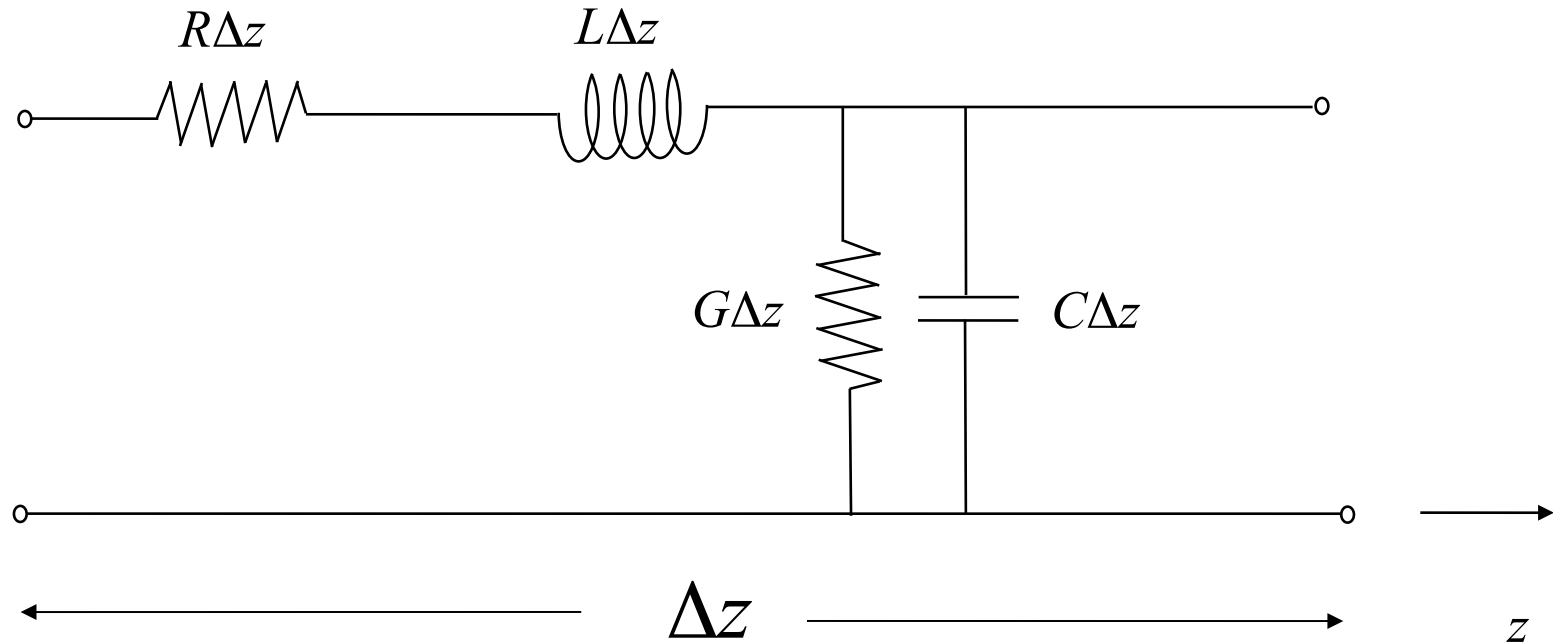


This applies for transmission lines

Jean-Baptiste-Joseph Fourier

Telegrapher's Equations

Transmission-line model for a small length of transmission line:



C = capacitance/length [F/m]

L = inductance/length [H/m]

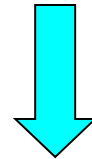
R = resistance/length [Ω /m]

G = conductance/length [S/m]

Telegrapher's Equations (cont.)

- Take the derivative of the first TE with respect to z .
- Substitute in from the second TE.

$$\begin{aligned}\frac{\partial V}{\partial z} &= -RI - L\frac{\partial I}{\partial t} \\ \frac{\partial I}{\partial z} &= -GV - C\frac{\partial V}{\partial t}\end{aligned}$$



$$\frac{\partial^2 V}{\partial z^2} - (RG)V - (RC + LG)\frac{\partial V}{\partial t} - LC\left(\frac{\partial^2 V}{\partial t^2}\right) = 0$$

Recall: There is no exact solution in the time domain, for the lossy case.

Frequency Domain

To convert to the phasor domain, we use: $\frac{\partial}{\partial t} \rightarrow j\omega$

$$\frac{\partial V}{\partial z} = -RI - L \frac{\partial I}{\partial t} \quad \longrightarrow \quad \frac{\partial V}{\partial z} = -(R + j\omega L)I$$

$$\frac{\partial I}{\partial z} = -GV - C \frac{\partial V}{\partial t} \quad \longrightarrow \quad \frac{\partial I}{\partial z} = -(G + j\omega C)V$$

From the above equations, we can easily obtain

$$\frac{d^2 V}{dz^2} = (R + j\omega L)(G + j\omega C)V$$

Frequency Domain (cont.)

$$\frac{d^2 V}{dz^2} = (R + j\omega L)(G + j\omega C)V$$

Define

$$\gamma^2 = (R + j\omega L)(G + j\omega C)$$

Then

$$\frac{d^2 V}{dz^2} = \gamma^2 V$$

Solution:

$$V(z) = Ae^{-\gamma z} + Be^{+\gamma z}$$

Note: We have an exact solution, even for a lossy line!

Propagation Constant

Convention: we choose the (complex) square root to be the **principle branch**:

$$\gamma \equiv \sqrt{(R + j\omega L)(G + j\omega C)} \quad (\text{lossy case})$$

γ is called the **propagation constant**, with units of [1/m]

Principle branch of square root:

$$c = |c|e^{j\phi}$$

$$\begin{aligned}\sqrt{c} &= (|c|e^{j\phi})^{1/2} \\ &= \sqrt{|c|} e^{j(\phi/2)}\end{aligned}$$

$$-\pi < \phi \leq \pi$$

Note:

$$-\pi/2 < \phi/2 \leq \pi/2$$



$$\operatorname{Re} \sqrt{c} \geq 0$$

Propagation Constant (cont.)

$$\gamma = \sqrt{(R + j\omega L)(G + j\omega C)}$$

Denote:

$$\gamma = \alpha + j\beta$$

γ = propagation constant [1/m]
 α = attenuation constant [nepers/m]
 β = phase constant [radians/m]

Choosing the principle branch means that

$$\operatorname{Re} \gamma \geq 0 \quad \longrightarrow \quad \alpha \geq 0$$

Propagation Constant (cont.)

Physical interpretation of waves:

$$V^+(z) = A e^{-\gamma z} \quad (\text{forward traveling wave})$$

$$V^-(z) = B e^{+\gamma z} \quad (\text{backward traveling wave})$$

(This interpretation will be shown shortly.)

Forward traveling wave: $V^+(z) = A e^{-\alpha z} e^{-j\beta z}$

Forward Wave

Forward traveling wave: $V^+(z) = A e^{-\alpha z} e^{-j\beta z}$

Denote $A = |A| e^{j\phi}$

Then $V^+(z) = |A| e^{j\phi} e^{-\alpha z} e^{-j\beta z}$

In the time domain we have:

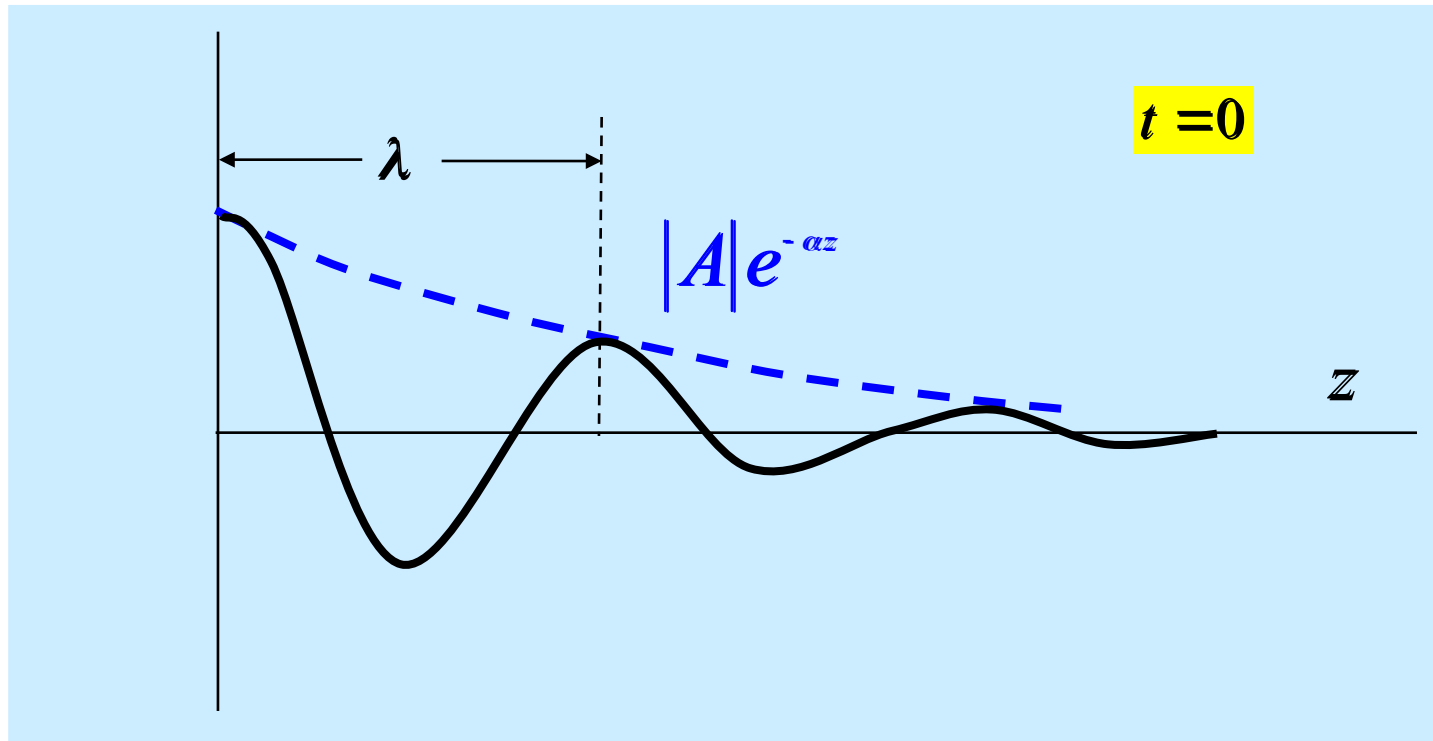
$$V^+(z, t) = \text{Re} \left\{ V^+(z) e^{j\omega t} \right\}$$

Hence we have $V^+(z, t) = |A| e^{-\alpha z} \cos(\omega t - \beta z + \phi)$

Forward Wave (cont.)

$$V^+(z,t) = |A| e^{-\alpha z} \cos(\omega t - \beta z + \phi)$$

Snapshot of Waveform:



The distance λ is the distance it takes for the waveform to “repeat” itself in meters.

$$\lambda = \text{wavelength}$$

Wavelength

The wave “repeats” (except for the amplitude decay) when:

$$\beta\lambda = 2\pi$$

Hence:

$$\beta = \frac{2\pi}{\lambda}$$

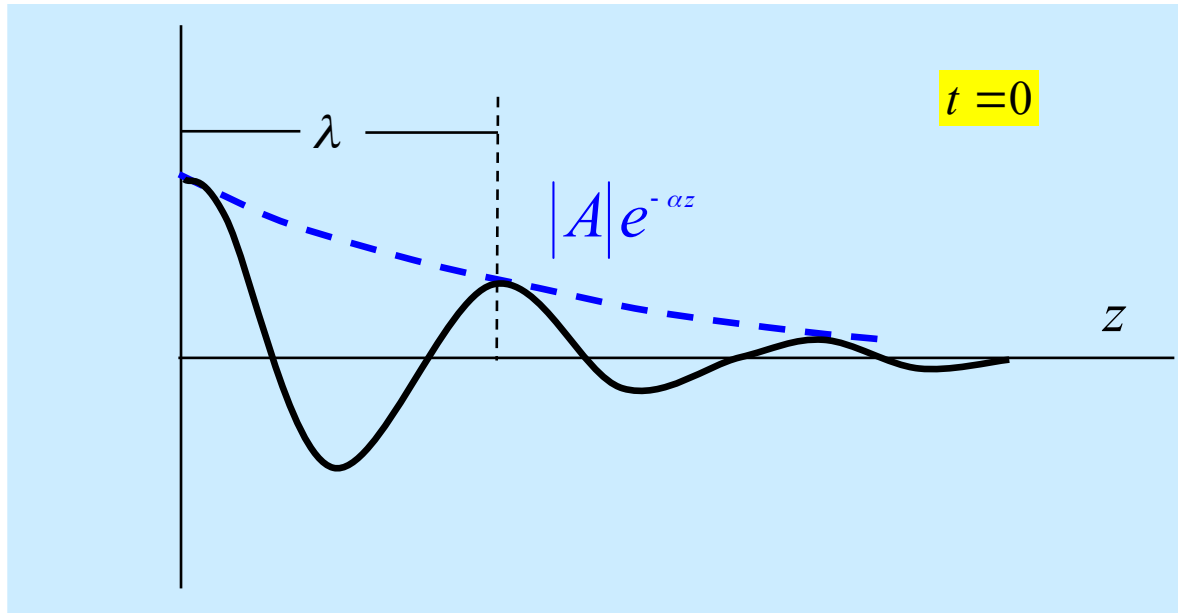
$$\lambda = \frac{2\pi}{\beta}$$

Attenuation Constant

The **attenuation constant** controls how fast the wave decays.

$$V^+(z, t) = |A| e^{-\alpha z} \cos(\omega t - \beta z + \phi)$$

$$\text{envelope} = |A| e^{-\alpha z}$$



Lossless Case

For a **lossless line**, we consider this as the limit of a lossy line, in the limit as the loss tends to zero:

$$\gamma = \sqrt{(R + j\omega L)(G + j\omega C)} = j\omega\sqrt{LC} \quad (\text{lossless case})$$

Hence we have that

$$\alpha = 0$$

$$\beta = \omega\sqrt{LC} = \omega\sqrt{\mu\epsilon}$$

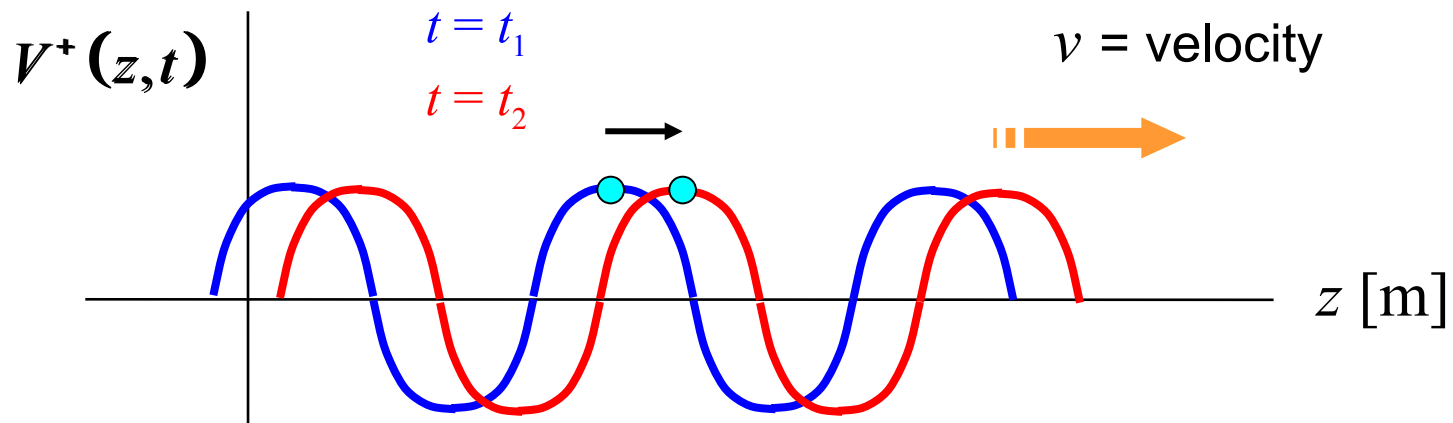
Note: $\alpha = 0$ for a lossless line.

Phase Velocity

The forward-traveling wave is moving in the positive z direction.

Consider a lossless transmission line for simplicity ($\alpha = 0$):

$$V^+(z, t) = |A| \cos(\omega t - \beta z + \phi)$$



Crest of wave: $\omega t - \beta z + \phi = 0$

Phase Velocity (cont.)

The **phase velocity** v_p is the velocity of a point on the wave, such as the crest.

$$\text{Set } \omega t - \beta z = \text{constant}$$

Take the derivative with respect to time: $\omega - \beta \frac{dz}{dt} = 0$

Hence

$$\frac{dz}{dt} = \frac{\omega}{\beta}$$

We thus have

$$v_p = \frac{\omega}{\beta}$$

Note: This result holds also for a lossy line.)

Phase Velocity (cont.)

Let's calculate the phase velocity for a **lossless line**:

$$v_p = \frac{\omega}{\beta} = \frac{\omega}{\omega\sqrt{LC}} = \frac{1}{\sqrt{LC}}$$

Also, we know that $LC = \mu\varepsilon = \frac{1}{c_d^2}$

Hence

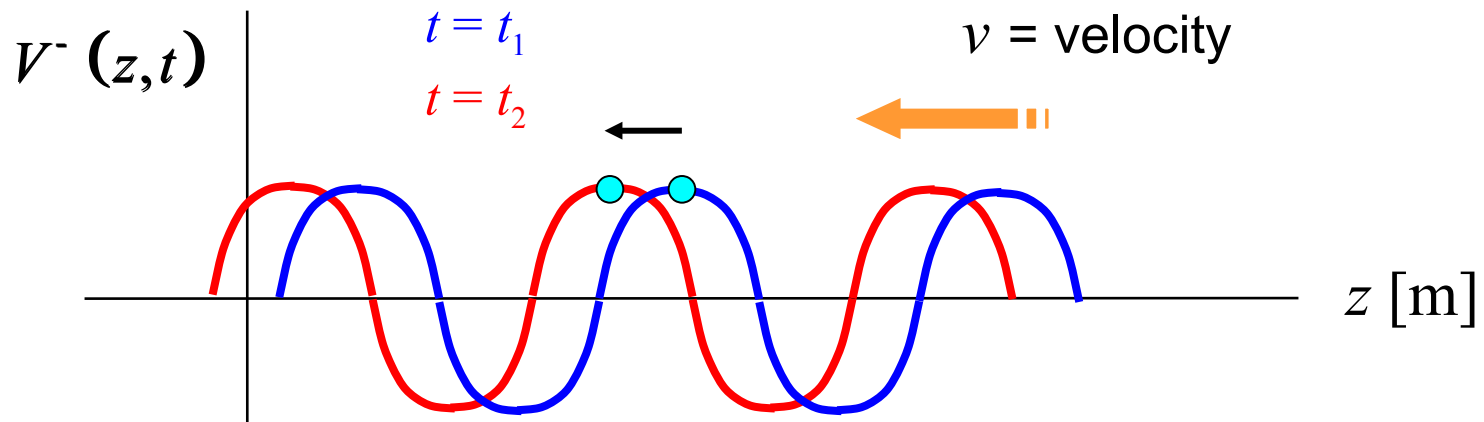
$$v_p = c_d$$

(lossless line)

Backward Traveling Wave

Let's now consider the backward-traveling wave (lossless, for simplicity):

$$V^-(z, t) = |A| \cos(\omega t + \beta z + \phi)$$



Attenuation in dB/m

$$V^+(z, t) = |A| e^{-\alpha z} \cos(\omega t - \beta z + \phi)$$

Attenuation in dB: $\text{dB} = 20 \log_{10} \left| \frac{V^+(z)}{V^+(0)} \right| = 20 \log_{10}(e^{-\alpha z})$

Use the following logarithm identity: $\log_{10} x = \frac{\ln x}{\ln 10}$

Therefore $\text{dB} = 20 \frac{\ln(e^{-\alpha z})}{\ln 10} = 20 \frac{(-\alpha z)}{\ln 10}$

Hence we have: $\text{attenuation} = \left(\frac{20}{\ln 10} \right) \alpha \quad [\text{dB/m}]$

Attenuation in dB/m (cont.)

Final attenuation formulas:

$$\text{attenuation} = \left(\frac{20}{\ln 10} \right) \alpha \quad [\text{dB/m}]$$

$$\text{attenuation} = (8.6859) \alpha \quad [\text{dB/m}]$$

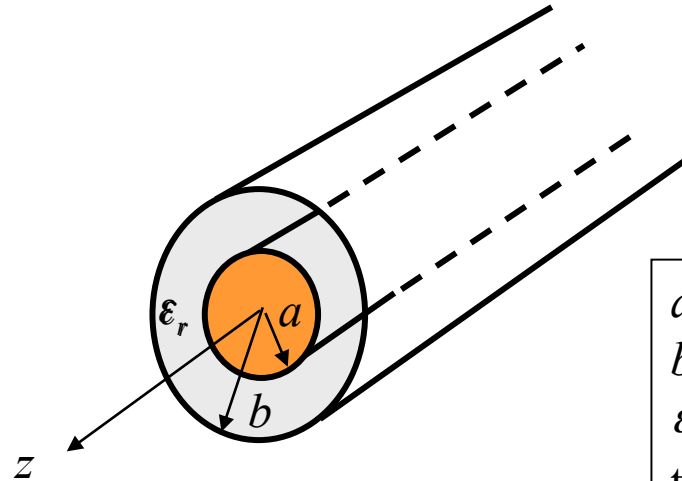
Example: Coaxial Cable

$$C = \frac{2\pi\epsilon_0\epsilon_r}{\ln\left(\frac{b}{a}\right)} \quad [\text{F/m}]$$

$$L = \frac{\mu_0}{2\pi} \ln\left(\frac{b}{a}\right) \quad [\text{H/m}]$$

$$G = \frac{2\pi\sigma_d}{\ln\left(\frac{b}{a}\right)} \quad [\text{S/m}]$$

$$R = \left(\frac{1}{2\pi a \sigma_{ma} \delta_{ma}} + \frac{1}{2\pi b \sigma_{mb} \delta_{mb}} \right) \quad [\Omega/\text{m}]$$



Copper conductors
(nonmagnetic: $\mu = \mu_0$)

$$\begin{aligned} a &= 0.5 \text{ [mm]} \\ b &= 3.2 \text{ [mm]} \\ \epsilon_r &= 2.2 \\ \tan\delta_d &= 0.001 \\ \sigma_{ma} = \sigma_{mb} &= 5.8 \times 10^7 \text{ [S/m]} \\ f &= 500 \text{ [MHz] (UHF)} \end{aligned}$$

$$\delta_m = \sqrt{\frac{2}{\omega\mu\sigma_m}} \quad (\text{skin depth})$$

Example: Coaxial Cable (cont.)

Ignore loss for calculating Z_0 :

$$Z_0 = \sqrt{\frac{L}{C}} = \frac{\eta_0}{2\pi\sqrt{\epsilon_r}} \ln\left(\frac{b}{a}\right)$$

$$Z_0 = 75 \text{ } [\Omega]$$

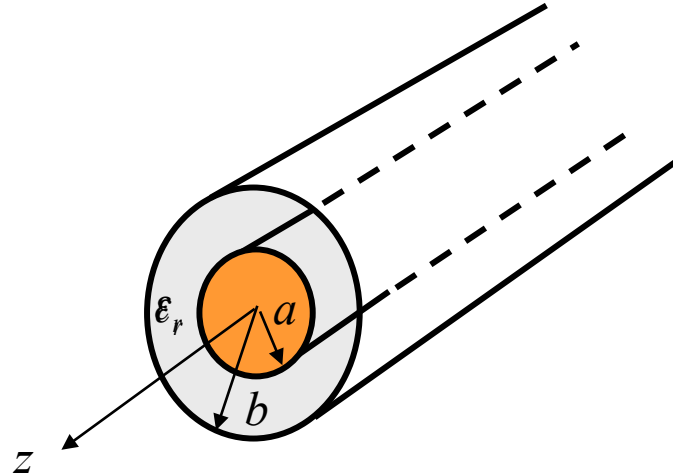
$$\delta_m = \sqrt{\frac{2}{\omega\mu\sigma_m}}$$

$$\delta_m = 2.955 \times 10^{-6} \text{ [m]}$$

$$\tan \delta_d \equiv \frac{\sigma_d}{\omega\epsilon} = \frac{\sigma_d}{\omega\epsilon_0\epsilon_r}$$

$$\sigma_d = (\omega\epsilon_0\epsilon_r) \tan \delta_d$$

$$\sigma_d = 6.12 \times 10^{-5} \text{ [S/m]}$$



$$a = 0.5 \text{ [mm]}$$

$$b = 3.2 \text{ [mm]}$$

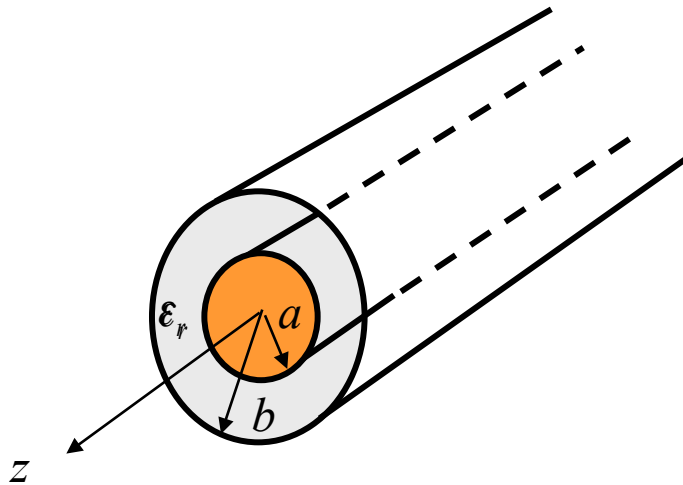
$$\epsilon_r = 2.2$$

$$\tan \delta_d = 0.001$$

$$\sigma_{ma} = \sigma_{mb} = 5.8 \times 10^7 \text{ [S/m]}$$

$$f = 500 \text{ [MHz] (UHF)}$$

Example: Coaxial Cable (cont.)



$$a = 0.5 \text{ [mm]}$$

$$b = 3.2 \text{ [mm]}$$

$$\epsilon_r = 2.2$$

$$\tan \delta_d = 0.001$$

$$\sigma_{ma} = \sigma_{mb} = 5.8 \times 10^7 \text{ [S/m]}$$

$$f = 500 \text{ [MHz] (UHF)}$$

$$R = 2.147 \text{ } [\Omega/\text{m}]$$

$$L = 3.713 \text{ E-07 [H/m]}$$

$$G = 2.071 \text{ E-04 [S/m]}$$

$$C = 6.593 \text{ E-11 [F/m]}$$

$$\gamma = \sqrt{(R + j\omega L)(G + j\omega C)}$$

$$\gamma = 0.022 + j(15.543) \text{ [1/m]}$$

$$\alpha = 0.022 \text{ [nepers/m]}$$

$$\beta = 15.543 \text{ [rad/m]}$$

$$\lambda = 0.404 \text{ [m]}$$

$$\text{attenuation} = 0.191 \text{ [dB/m]}$$

Current

Use the first Telegrapher equation:

$$\frac{\partial V}{\partial z} = -RI - L \frac{\partial I}{\partial t}$$

$$\Downarrow \quad \frac{\partial}{\partial t} \rightarrow j\omega$$

$$\frac{\partial V}{\partial z} = -RI - j\omega LI$$

Next, use

$$V(z) = Ae^{-\gamma z} + Be^{+\gamma z}$$

so

$$\frac{\partial V(z)}{\partial z} = -\gamma [Ae^{-\gamma z} - Be^{+\gamma z}]$$

Current (cont.)

Hence we have

$$-\gamma \left[A e^{-\gamma z} - B e^{+\gamma z} \right] = -RI - j\omega LI$$

Solving for the phasor current I, we have

$$\begin{aligned} I &= \left(\frac{\gamma}{R + j\omega L} \right) \left[A e^{-\gamma z} - B e^{+\gamma z} \right] \\ &= \left(\frac{\sqrt{(R + j\omega L)(G + j\omega C)}}{R + j\omega L} \right) \left[A e^{-\gamma z} - B e^{+\gamma z} \right] \\ &= \sqrt{\frac{G + j\omega C}{R + j\omega L}} \left[A e^{-\gamma z} - B e^{+\gamma z} \right] \end{aligned}$$

Characteristic Impedance

Define the (complex) characteristic impedance Z_0 :

$$Z_0 \equiv \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

Lossless:

$$Z_0 = \sqrt{\frac{L}{C}}$$

Then we have

$$V(z) = Ae^{-\gamma z} + Be^{+\gamma z}$$

$$I(z) = \left(\frac{1}{Z_0} \right) [Ae^{-\gamma z} - Be^{+\gamma z}]$$